

Temporal clustering of global $M \geq 6.5$ earthquakes is consistent with ETAS: a hold-out validation

Detector uses spatial gates; validation temporal-only

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Abstract.

We analyze an **analysis catalog of 4,267 unique $M \geq 6.5$ events** (modern window 1973–2026: 2,041). Baiesi-Paczuski η detector yields **27 algorithmic candidates**. **Catalog-calibrated temporal ETAS: $p_{\text{ETAS}}=1.0$** (in-sample null; full numbers — Sec. 4.1). The title refers to **detection geometry** ($\eta + GC > 1500$ km gates); **inferential tests are temporal-only**. Spatial component not modeled. Pre-1900 NOAA — Supplementary S3.

Keywords: global seismicity; seismic series; earthquake clustering; heuristic metric with tectonic hint; Baiesi-Paczuski metric; ETAS validation; Monte Carlo; paleoseismology; remote triggering

1. INTRODUCTION

Large earthquakes are not independent in time. Michael [2011] and Shearer & Stark [2012] found no anomalous global $M \geq 7$ clustering beyond standard null models, but neither tested multi-regional η -linkage at $M \geq 6.5$ with catalog-calibrated ETAS validation.

Two-phase design. (1) **Detection phase** — spatiotemporal *candidate search*: Baiesi-Paczuski η NN forest plus sliding-window detector (mean GC > 1500 km, $N \geq 4$, $M \geq 6.5$). (2) **Validation phase** — *temporal-only* catalog-calibrated ETAS MLE plus hold-out; does **not** test spatial linkage (Ogata 1998 spatial kernel — future work).

Objective. Temporal ETAS falsification plus a reproducible $M \geq 6.5$ pipeline (1973–2026). **Scope.** Phase (1) yields detector candidates; phase (2) reports temporal excess vs catalog-calibrated ETAS only. Bird (2003) — preliminary idea, not validated (Supplementary S1; excluded from primary pipeline).

§1.1 Research question. Do physically meaningful multi-regional global series exist in $M \geq 6.5$ (1973–2026)? (a) Permutation (diagnostic): Poisson times rejected — expected with aftershocks (Methods). (b) ETAS MLE → see Sec. 4.1. (c) Global-series hypothesis: **not tested (spatial ETAS open)**. (d) WLS (Supplementary S2): coupling illustration, not primary null.

2. DATA AND METHODS

2.1 Catalog compilation

Magnitude notation. Catalog thresholds and detector gates use $M \geq 6.5$ (USGS ComCat, predominantly M_w); individual events are cited with catalog type where relevant (e.g. M_w 7.3).

Three catalogs were merged: **USGS ComCat** (1900–2026, primary instrumental); **ISC Bulletin** (relocated hypocenters); and **NOAA NGDC** (historical and paleoseismic records pre-1900, 47 events). Duplicates were merged using ± 30 days and ≤ 50 km tolerance; source priority: ISC > USGS > NOAA. After deduplication, the catalog contains **4,267** unique $M \geq 6.5$ events (from **4,418** CSV rows; ~ 151 $M < 6.5$ excluded from clustering). USGS ComCat: **2,088** raw (1973–2026) → **2,041** after merge/ISC. GK: 24 aftershocks (2,017/2,041); ZBZ: 1 dependent (2,040/2,041). *quality_score* is metadata, not an inclusion filter. **Primary analysis set:** events 1900–2026 (4,218); 47 pre-1900 records remain in CSV (provenance) but are excluded from the primary detector pipeline and ETAS calibration window (1973–2026 only) — see Supplementary S3. **Final catalog:** 4,267 $M \geq 6.5$ events (4,418 CSV rows).

2.1.1 Catalog completeness

Maximum-curvature analysis yields $M_c = 6.55$. Maximum-likelihood b-value from 1,688 events above M_c :

$$b = 0.911 \pm 0.018 \text{ (n = 1,688 events)}$$

2.2 Tectonic heuristic (deprecated)

The Bird (2003) tectonic-path heuristic is **excluded from the primary pipeline; no synthetic benchmark** in this work; comparison deferred to supplementary material and future work. Primary analysis uses **great-circle distance only** (see paper/supplementary.md §S1).

$$\eta_{ij} = t_{ij} * r_{ij}^{1.6} * 10^{(-b * m_i)}$$

*b=1.0 — deliberate Baiesi & Paczuski (2004) simplification; catalog $b=0.911 \pm 0.018$ for M_c /completeness and MC null only — **not** in the η formula. Full pipeline at $b=0.911$ (run_sensitivity_b0911_full.py): $N=27$, Jaccard=1.0; **8.2%** upstream cluster-label mismatch — detector gates dominate. Changing b re-runs upstream identify_clusters() labels only, not global_series() gates (sensitivity_b0911_full_pipeline.json).*

2.3 Analysis pipeline

Raw catalogs → dedup → exclude $M < 6.5$ (~151 NOAA) → GK declustering (primary) → η NN forest → sliding windows → merge overlapping → series criteria → MC/ETAS/FDR. GK: 24 aftershocks (2,017/2,041); ZBZ: 1 dependent (2,040/2,041) — different algorithms (GK: magnitude-dependent windows; ZBZ: η NN + KDE threshold). At global $M \geq 6.5$ scale ZBZ is permissive (sparse events → high η). GK is sole primary for inference; ZBZ sensitivity only, does not replace GK. If ZBZ were primary, $N=27$ likely unchanged (untested).

Why GK-ZBZ counts differ: not contradictory methods — GK removes short-range fore/aftershocks via fixed windows; ZBZ classifies only 1 event with exceptionally low η at global catalog scale. Under fixed gates GK/ZBZ/none all yield $N=27$ (sensitivity_declustering.json); different mainshock labels (24 vs 1) could matter for cluster analysis.

Detection algorithm (§3.3). GK WINDOWS, $T(M)/R(M)$ interpolation, magnitude-descending; aftershocks $[0, T]$, foreshocks $[-T/2, 0]$; haversine. find_nearest_neighbor: causal argmin η , $b=1.0$, GC km. identify_clusters: Union-Find, η_0 KDE. global_series: used[] mask, window $[t_i, t_i + \Delta t]$, mean GC > 1500 km; merge 142→47. **Gates:** $N \geq 4$; $M \geq 6.5$; mean pairwise GC > 1500 km. Flinn-Engdahl zone count — diagnostic only. **Permutation diagnostic (not primary validation):** $p=0.0001$ (1/10,001), $z=-6.17$; rejects Poisson event times only (Ogata 1988). BH post-hoc — exploratory only.

2.4 Primary ETAS null

Primary ETAS null uses catalog-calibrated temporal MLE on GK mainshocks (calibrate_etas_mle.py). Hold-out: train 1973–2000 (1024 GK mainshocks), hold-out

2001–2026 (1010 events); MLE **train only**, parameters fixed without retraining (calibrate_etas_holdout.py; N_obs=13, p=1.0). WLS — Supplementary S2 only.

2.5 Statistical validation

Permutation statement: $p=0.0001$ rejects **Poisson event times only** (Ogata 1988); does not confirm global series. ETAS — Sec. 4.1. BH post-hoc — not discovery.

3. RESULTS

3.1 Identified series

Full historical analysis yields **47 detector candidates** (not validated global series): 27 modern, 15 early instrumental (quality caveat), 5 historical candidates (not significant, $p=0.46$). 142 cluster candidates before filtering. Primary validation — ETAS table below only.

Serie s ID	Ev ent s	Re gio ns	M ma x	Years	Mean lat°	Mean lon°	Notes
1905–1910	193	43	8.8	1905–1910	−60...72	−180...180	Early instrumental; incomplete before ~1960
S047	53	5	8.0	1982–2024	−21.5...18.9	−175.5...155.2	Western Pacific
S170	46	12	9.1	2002–2023	−14.3...33.8	−113.1...167.3	Sumatra 2004 (M 9.1)
S095	25	4	7.9	1989–2017	−8.1...16.5	120.4...146.4	Western Pacific arc
S116	22	5	8.2	1993–2021	−31.7...10.8	−179.5...179.4	South Pacific

Table 1. Raw detector candidates — NOT validated series; illustrative only.

3.2 ETAS validation (Sec. 4.1 canonical)

Split	N_obs	mean	p_ETAS
In-sample (1973–2026)	27	27.0	1.0
Train-calibrated hold-out (2001–2026)	13	13.0	1.0

Hold-out — partial out-of-time check, not spatial validation. **Multiplicity correction (142 windows):** Bonferroni $0.05/142 \approx 0.00035$ — **0** significant windows/series; BH FDR $q=0.05$ — **0**. No claim of temporal anomalies after correction. Primary inference: **consistency with in-sample temporal ETAS** ($p_{\text{ETAS}}=1.0$). Post-hoc BH on 47 merged — exploratory only.

3.3 Spatial-temporal distribution

Elevated detector candidate frequency (not validated physical series) occurs in 1952–1965 and 2002–2016 (post-Sumatra period). Spatially, clusters concentrate along the circum-Pacific belt (Kamchatka, Kuril Islands, Japan, Tonga, Indonesia). Tectonic diagnostic: median $\Delta \log_{10} \eta = +0.28$ on random pairs (mostly $1.5\times$ GC fallback).

3.4 Consolidated sensitivity table (modern window)

Parameter	Setting	N_series
GC gate	1000 km	27
GC gate	1500 km (baseline)	27
GC gate	2000 km	27
Window	1 yr	53
Window	2 yr (baseline)	27
Window	5 yr	11
Window	10 yr	6
b in η	1.0 (BP 2004)	27
b in η	0.911 (catalog)	27
b overlap (full pipeline)	Jaccard=1.0; upstream 8.2%‡	27
Declustering	GK / ZBZ / none	27 / 27 / 27
min_events (strict)	5 / 6 / 8	27 / 27 / 27
Catalog	GK mainshocks only	27
$\eta_0 \pm 20\%$	not_applied	—

Table 2. *N_series* sensitivity under fixed detector gates (mean GC > 1500 km, $N \geq 4$). Sources: *sensitivity_*.json*. ‡**8.2%** = 165/2017 GK mainshocks whose *identify_clusters()* cluster label changes when *b* goes from 1.0 to 0.911 (full pipeline; *sensitivity_b0911_full_pipeline.json*).

The row **b overlap (full pipeline)** reports a full re-run $GK \rightarrow identify_clusters() \rightarrow global_series()$ at $b = 1.0$ vs 0.911. **Jaccard = 1.0** means identical **series event sets** across 27 detections --- *global_series()* does not use *b* in η . **8.2%** upstream is the fraction of GK mainshocks whose **η -cluster label** changes at *identify_clusters()*; this is a different level from downstream series membership, so $N = 27$ and Jaccard = 1.0 do **not** prove unchanged upstream structure.

N = 27 stability. At fixed gates the same event kernel survives merge $142 \rightarrow 47 \rightarrow 27$ under GK/ZBZ/none (Jaccard of event sets = 1.0; series membership Jaccard = 0.32 for ZBZ/none). Window width dominates (53 at 1 yr, 11 at 5 yr). Detector-artifact stability, not a physical invariant “core 27.”

4. DISCUSSION AND CONCLUSIONS

We found **no evidence of temporal anomalies** per Bonferroni-corrected window tests; the **spatial question remains open**. **Temporal ETAS:** consistent with catalog-calibrated null ($p_{ETAS}=1.0$). Spatial linkage not tested. Bird/WLS/pre-1900 — paper/supplementary.md §S1–S3.

Supplementary material. Bird (2003), WLS negative control, pre-1900 NOAA: paper/supplementary.md (§S1–S3).

Limitation	Affected step	Impact on main conclusion
η_0 unverified at global scale	GK/ZBZ identify_clusters	KDE valley $\approx 7.1 \times 10^{-6}$; global_series skips $\eta_0 \rightarrow N=27$; mis-specified η_0 shifts GK/ZBZ labels only
$b = 1.0$ vs 0.911 in η	η upstream	$N_{\text{series}} = 27$ both; 8.2% upstream label mismatch (full pipeline)
No spatial Ogata MLE	ETAS null	Temporal MLE primary; spatial MLE future work
142 windows + merge	Detector	Main source of liberalness
Calibrated $p_{\text{ETAS}} = 1.0$	ETAS test	Consistent with null; supports negative conclusion

Table 3. Limitation $\rightarrow$ affected step $\rightarrow$ impact on main conclusion (Sec. 5.6).

Impact analysis. $N = 27$ is computed by global_series() without η_0 filtering; full pipeline $b = 1.0$ vs 0.911 leaves $N_{\text{series}} = 27$, Jaccard = 1.0, but 8.2% upstream label mismatch. 142 sliding windows dominate detector liberalness.

DATA AND CODE AVAILABILITY

Data and code: github.com/marshalkin-ux/paleoseismic-clustering (docs/data_availability.md). External DOI deposition (Zenodo) deferred — GitHub only.

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